ONE-PRIME RAMIFIED REDUCTIVE P-ADIC GALOIS REPRESENTATIONS WITH LARGE IMAGE VIA RELATIVE DEFORMATION LIFTING

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

Let p be a prime and let $\mathbf{G}$ be a reductive group. The problem of constructing Galois representations with large image in $\mathbf{G}(\mathbb{Z}_p)$ may be viewed as a p-adic analogue of the inverse Galois problem. In the classical inverse Galois problem one asks which finite groups occur as Galois groups over $\mathbb{Q}$. Here the target is instead a compact p-adic group, and one asks for continuous representations of the absolute Galois group whose image is open, or at least contains a principal congruence subgroup of the derived group. A natural refinement is to prescribe, or minimize, the set of primes at which the representation is allowed to ramify.

There are many natural sources of Galois representations in arithmetic geometry. When the dimension is two, representations with large image arise abundantly from elliptic curves and modular forms. In higher dimensions one may consider Galois representations associated with motives, automorphic forms, or the Tate modules of abelian varieties. These geometric constructions, however, frequently impose additional structure on the image. For example, the representation on the Tate module of a polarized abelian variety is constrained by the Weil pairing. It is therefore difficult, by purely geometric means, to produce higher-dimensional representations into $\mathrm{GL}_n(\mathbb{Z}_p)$ whose images contain open subgroups of $\mathrm{SL}_n(\mathbb{Z}_p)$ and which at the same time satisfy very restrictive ramification conditions. This suggests a different point of view. Rather than insisting that the representation arise from a geometric or automorphic object, one can construct it directly by Galois-theoretic and deformation-theoretic methods.

The ramification problem considered in this paper is obtained by allowing ramification at only one finite prime. Let $\mathbb{Q}_{\{p\}}$ be the maximal algebraic extension of $\mathbb{Q}$ which is unramified at every finite prime different from p, and put $G_{\{p\}} := \mathrm{Gal}(\mathbb{Q}_{\{p\}}/\mathbb{Q})$. Thus a representation of $G_{\{p\}}$ is a representation of $G_{\mathbb{Q}}$ which is unramified at every finite prime $\ell \neq p$. The question is whether the very severe restriction on ramification is compatible with having a large p-adic image.

This problem originates in work of Greenberg [2]. Let $\bar{\chi}$ denote the mod-p cyclotomic character and let C be the mod-p class group of $\mathbb{Q}(\mu_p)$. The action of $\mathrm{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$ decomposes C into eigenspaces $C(\bar{\chi}^i)$. The prime p is regular precisely when $C = 0$. Vandiver's conjecture predicts the vanishing of the even eigenspaces, or equivalently of the p-primary part of the class group of the maximal totally real subfield of $\mathbb{Q}(\mu_p)$. Thus the arithmetic of the cyclotomic

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class group enters naturally into the attempt to construct one-prime Galois representations. A theorem of Shafarevich implies that when p is regular, the maximal pro-p extension of $\mathbb{Q}(\mu_p)$ unramified outside the primes above p has free pro-p Galois group of explicitly known rank. Greenberg exploits this freeness to construct representations $G_{\{p\}} \longrightarrow \mathrm{GL}_n(\mathbb{Z}_p)$ whose images contain an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$, provided that p is regular and $p \geq 4\lfloor n/2 \rfloor + 1$.

Motivated by Greenberg's construction, the author formulated in [10] the following basic problem: for a given prime p and $n > 1$, does there exist a continuous representation $G_{\{p\}} \to \mathrm{GL}_n(\mathbb{Z}_p)$ whose image contains an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$? More generally, how small can the ramification set of a p-adic representation with large image be?

The first deformation-theoretic treatment of this question was given in *loc.cit.* Instead of requiring the entire mod-p class group of $\mathbb{Q}(\mu_p)$ to vanish, one begins with a carefully chosen diagonal residual representation and imposes vanishing only on the particular cyclotomic eigenspaces which occur in its adjoint representation. If e_p denotes the index of irregularity, this gives infinitely many representations unramified outside p provided that p is sufficiently large in terms of n and e_p. The essential deformation-theoretic input is the vanishing $H^2(G_{\{p\}}, \mathrm{Ad}\,\bar{\rho}) = 0$ for the chosen residual representation $\bar{\rho}$. Once this unobstructedness has been established, the remaining part of the argument consists in modifying level lifts by suitable H^1 Galois cohomology classes and then proving that the resulting image contains a principal congruence subgroup.

The dependence on the irregularity index and on the size of p was substantially reduced in the author's subsequent paper [11]. There it is shown that for every $n > 1$ and every $p \geq 7$ there are infinitely many representations $G_{\{p\}} \to \mathrm{GL}_n(\mathbb{Z}_p)$ whose images contain an explicitly bounded principal congruence subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$, subject, when $p \equiv 1 \pmod 4$, to a weak form of Vandiver's conjecture.

There are several related developments which help place this question in context. Katz [3] constructs finitely ramified representations with large image by geometric methods and Hilbert irreducibility, although the resulting ramification set is not restricted to the single prime p. Maire [5] gives a different unconditional construction of representations with large image in $\mathrm{GL}_n(\mathbb{Z}_p)$. For $p \equiv 3 \pmod 4$ the representations may be taken to be unramified outside p, while for $p \equiv 1 \pmod 4$ his construction allows the additional finite prime 2. Maletto [6] develops a particularly explicit version of the residual method for GSp_4.

Of particular relevance here is Tang's extension of the deformation-theoretic construction to reductive groups [12]. Let $\mathbf{G}$ be split reductive with $\dim Z(\mathbf{G}) \leq 1$. Tang starts with a cyclotomic representation in a split maximal torus, decomposes the adjoint representation into its root spaces, introduces root directions by Galois cohomology classes, and uses the Chevalley commutator relations to obtain open image. In this sense, Tang identifies the natural root-theoretic form of the construction in [10]. In this construction, the residual representation is chosen so that the corresponding H^2 vanishes. This requires avoiding the cyclotomic class-group eigenspaces that may contribute to the obstruction and leads to a lower bound on p depending on the root datum and a bound for the irregularity index.

The aim of the present paper is to remove any regularity assumptions on p and prove results that do not assume any form of Vandiver's conjecture. The deformation-theoretic approach belongs to the Galois-cohomological lifting tradition initiated by Ramakrishna [9]. The argument used here is particularly reminiscent of the relative deformation philosophy of Fakhruddin–Khare–Patrikis [1]. The key advantage is that one no longer requires the obstruction group itself to vanish. As one passes between representations modulo successive powers of p, there are natural comparison maps between the corresponding adjoint modules and hence between their second cohomology groups. At sufficiently high levels these cohomological comparison maps

become injective. Thus an obstruction which vanishes after passing to the lower level must already vanish at the higher level, and this is enough to continue the lifting process even when the deformation problem is genuinely obstructed.

We now state the main theorem. We first recall explicitly the two root-theoretic conditions which enter its formulation. Let $\mathbf{T} \subset \mathbf{G}$ be a split maximal torus, let $\Phi = \Phi(\mathbf{G}, \mathbf{T})$ be the associated root system, and let $\Delta \subset \Phi$ be a set of simple roots. We say that $(\mathbf{G}, \mathbf{T}, \Delta)$ *admits an odd cocharacter* if there exists $\lambda_0 \in X_*(\mathbf{T})$ such that $\langle \alpha, \lambda_0 \rangle \equiv 1 \pmod 2$ for all $\alpha \in \Delta$. The odd-cocharacter condition is automatic for several standard classical groups. For the usual choices of maximal torus and simple roots, one can in fact choose a cocharacter whose pairing with every simple root is equal to 1. This holds for GL_n, SO_{2n+1}, SO_{2n}, and GSp_{2n}.

Fix a pinning and corresponding Chevalley root vectors X_α, $\alpha \in \Phi$. Whenever $\alpha, \beta, \alpha+\beta \in \Phi$, write

$$[X_\alpha, X_\beta] = N_{\alpha,\beta} X_{\alpha+\beta},$$

and for $\delta \in \Delta$ put $H_\delta = [X_\delta, X_{-\delta}]$. We say that an odd prime p is *Lie-good for* $\mathbf{G}$ if every nonzero structure constant $N_{\alpha,\beta}$ occurring in these root-generation relations is nonzero modulo p, and if the reductions of the root vectors X_α together with the simple coroot directions H_δ span $\mathrm{Lie}(\mathbf{G}^{\mathrm{der}}) \otimes_{\mathbb{Z}} \mathbb{F}_p$. For a fixed split integral root datum it excludes only finitely many primes, and for $\mathbf{G} = \mathrm{GL}_n$ it is automatic for every prime.

Theorem 1.1. *Let $\mathbf{G}/\mathbb{Z}$ be a split connected reductive group with $\dim Z(\mathbf{G}) \leq 1$. Fix a split maximal torus $\mathbf{T} \subset \mathbf{G}$ and a set of simple roots Δ. Assume that $(\mathbf{G}, \mathbf{T}, \Delta)$ admits an odd cocharacter. Let p be an odd prime which is Lie-good for $\mathbf{G}$. Then there exist infinitely many continuous representations $\rho : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}_p)$ whose images contain a principal congruence subgroup of $\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$. If $\dim Z(\mathbf{G}) = 1$, the representations may be chosen so that their images in $(\mathbf{G}/\mathbf{G}^{\mathrm{der}})(\mathbb{Z}_p)$ are open; in particular their images are open in $\mathbf{G}(\mathbb{Z}_p)$.*

We obtain the following consequence for GL_n.

Corollary 1.2. *Let p be any odd prime and let $n > 1$. Then there exist infinitely many continuous representations $\rho : G_{\{p\}} \longrightarrow \mathrm{GL}_n(\mathbb{Z}_p)$ unramified outside $\{p, \infty\}$ whose images contain a principal congruence subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$. The determinant may be chosen to have open image, and hence the resulting representations have open image in $\mathrm{GL}_n(\mathbb{Z}_p)$.*

2. Preliminaries

In this section we discuss preliminary notions as well as set up notation used throughout the article.

Let $\overline{\mathbb{Q}}$ be a fixed algebraic closure of $\mathbb{Q}$ and let $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. For a finite set S of rational primes, let $\mathbb{Q}_S$ be the maximal algebraic extension of $\mathbb{Q}$ contained in $\overline{\mathbb{Q}}$ which is unramified at every finite prime outside S, and set $G_S = \mathrm{Gal}(\mathbb{Q}_S/\mathbb{Q})$. A continuous representation of $G_{\mathbb{Q}}$ which is unramified at every finite prime $\ell \neq p$ factors uniquely through $G_{\{p\}}$.

For every rational prime ℓ choose an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ and thereby a decomposition subgroup $G_\ell \subset G_{\mathbb{Q}}$. We write I_ℓ for its inertia subgroup. The p-adic cyclotomic character is

$$\chi : G_{\{p\}} \longrightarrow \mathbb{Z}_p^\times,$$

and $\overline{\chi}$ denotes its reduction modulo p. Set $F_\infty = \mathbb{Q}(\mu_{p^\infty})$ and $\Gamma = \mathrm{Gal}(F_\infty/\mathbb{Q})$. Since p is odd, we have that $\Gamma = \Delta_p \times \Gamma_1$, where $\Delta_p \simeq (\mathbb{Z}/p\mathbb{Z})^\times$ and $\Gamma_1 \simeq 1 + p\mathbb{Z}_p \simeq \mathbb{Z}_p$. Let ω be the Teichmuller character of Δ_p. Choose a topological generator γ of Γ_1 and set $u = \chi(\gamma)$. We write $\Lambda = \mathbb{Z}_p[[\Gamma_1]] \simeq \mathbb{Z}_p[[T]]$, with $\gamma^{-1} = 1 + T$. Finally set $H = \mathrm{Gal}(\mathbb{Q}_{\{p\}}/F_\infty)$.

Let $\mathbf{G}/\mathbb{Z}$ be a split connected reductive group and let $\mathbf{T} \subset \mathbf{G}$ be a split maximal torus. Fix a Borel subgroup containing $\mathbf{T}$ and let $\Phi = \Phi(\mathbf{G}, \mathbf{T})$, $\Delta \subset \Phi$ and $\Phi^+ \subset \Phi$ be the associated root system, simple roots, and positive roots respectively. Let $\mathbf{G}^{\mathrm{der}}$ be the derived subgroup. We set $\mathfrak{g}_{\mathbb{Z}_p} = \mathrm{Lie}(\mathbf{G}) \otimes \mathbb{Z}_p$, $\mathfrak{t}_{\mathbb{Z}_p} = \mathrm{Lie}(\mathbf{T}) \otimes \mathbb{Z}_p$ and $\mathfrak{g}^{\mathrm{der}}_{\mathbb{Z}_p} = \mathrm{Lie}(\mathbf{G}^{\mathrm{der}}) \otimes \mathbb{Z}_p$. For a coefficient ring R we use the corresponding subscript R.

Choose a pinning. For every root $\alpha \in \Phi$ this gives a root vector X_α spanning a free rank-one root submodule. The integral root decomposition is

$$\mathfrak{g}_{\mathbb{Z}_p} = \mathfrak{t}_{\mathbb{Z}_p} \oplus \bigoplus_{\alpha \in \Phi} \mathbb{Z}_p X_\alpha. \tag{2.1}$$

If $\alpha + \beta \in \Phi$, the Chevalley relation has the form

$$[X_\alpha, X_\beta] = N_{\alpha,\beta} X_{\alpha+\beta}, \tag{2.2}$$

where $N_{\alpha,\beta} \in \mathbb{Z} \setminus \{0\}$, and

$$[X_\alpha, X_{-\alpha}] = H_\alpha \in \mathfrak{t}_{\mathbb{Z}_p} \cap \mathfrak{g}^{\mathrm{der}}_{\mathbb{Z}_p}. \tag{2.3}$$

For $\alpha = \sum_{\delta \in \Delta} n_\delta \delta \in \Phi^+$ define

$$\mathrm{ht}(\alpha) = \sum_{\delta \in \Delta} n_\delta,$$

and set $\mathrm{ht}(-\alpha) = -\mathrm{ht}(\alpha)$. Let

$$X_*(\mathbf{T}) := \mathrm{Hom}_{\mathrm{alg.grp.}}(\mathbf{G}_m, \mathbf{T})$$

denote the cocharacter lattice of the split maximal torus $\mathbf{T}$, and let

$$X^*(\mathbf{T}) := \mathrm{Hom}_{\mathrm{alg.grp.}}(\mathbf{T}, \mathbf{G}_m)$$

denote its character lattice. There is a natural perfect pairing

$$\langle\ ,\ \rangle : X^*(\mathbf{T}) \times X_*(\mathbf{T}) \longrightarrow \mathbb{Z}$$

characterized by the identity

$$\alpha \circ \lambda(t) = t^{\langle \alpha, \lambda \rangle}$$

for $\alpha \in X^*(\mathbf{T})$ and $\lambda \in X_*(\mathbf{T})$. In particular, each root $\alpha \in \Phi(\mathbf{G}, \mathbf{T}) \subset X^*(\mathbf{T})$ may be paired with a cocharacter.

We write

$$\mathbb{Z}^\Delta := \prod_{\alpha \in \Delta} \mathbb{Z},$$

whose coordinates are indexed by the simple roots, and consider the homomorphism

$$\vartheta : X_*(\mathbf{T}) \longrightarrow \mathbb{Z}^\Delta$$

defined by $\lambda \longmapsto \big(\langle \alpha, \lambda \rangle\big)_{\alpha \in \Delta}$. Thus an odd cocharacter is precisely a cocharacter λ for which every coordinate of $\vartheta(\lambda)$ is odd. Equivalently, after reducing the above map modulo 2, its image is the vector

$$(1, \ldots, 1) \in (\mathbb{Z}/2\mathbb{Z})^\Delta.$$

Let $\lambda \in X_*(\mathbf{T})$ be a cocharacter. Composing with the cyclotomic character gives

$$\rho_\lambda = \lambda \circ \chi : G_{\{p\}} \longrightarrow \mathbf{T}(\mathbb{Z}_p) \subset \mathbf{G}(\mathbb{Z}_p). \tag{2.4}$$

For a root α put

$$d_\alpha = \langle \alpha, \lambda \rangle \in \mathbb{Z}.$$

Conjugation by $\lambda(t)$ on the root line is multiplication by t^{d_α}. Consequently

$$\operatorname{Ad}\rho_\lambda(\mathfrak{g}_{\mathbb{Z}_p}) = \mathfrak{t}_{\mathbb{Z}_p} \oplus \bigoplus_{\alpha\in\Phi} \mathbb{Z}_p(d_\alpha)X_\alpha, \tag{2.5}$$

where $G_{\{p\}}$ acts trivially on $\mathfrak{t}_{\mathbb{Z}_p}$ and through χ^{d_α} on the root line. This decomposition is the basic bridge between the root datum and the cyclotomic cohomology considered in Section 4.

When $\dim Z(\mathbf{G}) = 1$, put $\mathbf{S} = \mathbf{G}/\mathbf{G}^{\mathrm{der}}$. Since $\mathbf{G}$ is split, $\mathbf{S}$ is a one-dimensional split torus. The image of λ in $X_*(\mathbf{S})$ will be denoted λ^{ab}. When $\lambda^{\mathrm{ab}} \neq 0$, the composition

$$G_{\{p\}} \xrightarrow{\rho_\lambda} \mathbf{G}(\mathbb{Z}_p) \longrightarrow \mathbf{S}(\mathbb{Z}_p)$$

is a nonzero integral power of χ and hence has open image.

We now make precise the second root-theoretic condition appearing in the main theorem. Recall that $\Phi = \Phi(\mathbf{G},\mathbf{T})$ is the root system of $\mathbf{G}$ with respect to the fixed split maximal torus $\mathbf{T}$, and that $\Delta \subset \Phi$ is the chosen set of simple roots. Put $\mathfrak{g}_{\mathbb{Z}} := \mathrm{Lie}(\mathbf{G})$ and $\mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}} := \mathrm{Lie}(\mathbf{G}^{\mathrm{der}})$, where $\mathbf{G}^{\mathrm{der}}$ denotes the derived subgroup of $\mathbf{G}$. Thus $\mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}}$ is the integral Lie algebra of the semisimple group $\mathbf{G}^{\mathrm{der}}$.

Fix the pinning chosen above. For every root $\alpha \in \Phi$, let $X_\alpha \in \mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}}$ denote the corresponding Chevalley root vector. The root submodule associated with α is the free rank-one $\mathbb{Z}$-module $\mathbb{Z}X_\alpha$. For each simple root $\delta \in \Delta$, put

$$H_\delta := [X_\delta, X_{-\delta}] \in \mathrm{Lie}(\mathbf{T}) \cap \mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}}.$$

Thus the vectors X_α give the root directions in the derived Lie algebra, while the elements H_δ give the simple coroot directions.

Define

$$\mathcal{L}_{\mathrm{root}} := \langle X_\alpha\ (\alpha\in\Phi),\ H_\delta\ (\delta\in\Delta)\rangle_{\mathbb{Z}} \subseteq \mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}}.$$

After tensoring with $\mathbb{Q}$, the root vectors together with the simple coroot directions span the full semisimple Lie algebra. Hence

$$\mathcal{L}_{\mathrm{root}} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}} \otimes_{\mathbb{Z}} \mathbb{Q}.$$

It follows that $\mathcal{L}_{\mathrm{root}}$ has finite index in $\mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}}$. We denote this index by

$$c_{\mathbf{G}} := \left[\mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}} : \mathcal{L}_{\mathrm{root}}\right].$$

If $p \nmid c_{\mathbf{G}}$, reduction modulo p shows that the images of the root vectors X_α and the simple coroot directions H_δ span $\mathfrak{g}_{\mathbb{F}_p}^{\mathrm{der}} := \mathfrak{g}_{\mathbb{Z}}^{\mathrm{der}} \otimes_{\mathbb{Z}} \mathbb{F}_p$. This is one part of the Lie-good condition.

The other part concerns the Chevalley structure constants. Whenever $\alpha,\beta,\alpha+\beta \in \Phi$, one has

$$[X_\alpha, X_\beta] = N_{\alpha,\beta}X_{\alpha+\beta},$$

where $N_{\alpha,\beta} \in \mathbb{Z}\setminus\{0\}$. Set

$$N_{\mathbf{G}} := \prod_{\substack{\alpha,\beta\in\Phi\\ \alpha+\beta\in\Phi}} |N_{\alpha,\beta}|.$$

Thus $N_{\mathbf{G}}$ is a positive integer depending only on the chosen integral root datum. If $p \nmid c_{\mathbf{G}}N_{\mathbf{G}}$, then every Chevalley constant which occurs in the root-generation argument is nonzero modulo p, and the root and coroot directions span $\mathfrak{g}_{\mathbb{F}_p}^{\mathrm{der}}$. Consequently every odd prime $p \nmid c_{\mathbf{G}}N_{\mathbf{G}}$ is Lie-good for $\mathbf{G}$.

For $\mathbf{G} = \mathrm{GL}_n$, we have that $X_{e_i - e_j} = e_{ij}$. Whenever a root sum occurs the corresponding structure constant is ± 1; for example, $[e_{ij}, e_{jk}] = e_{ik}$. Moreover

$$\{e_{ij} : i \neq j\} \cup \{e_{ii} - e_{i+1,i+1} : 1 \leq i < n\}$$

is an integral basis of $\mathfrak{sl}_n$. Therefore $c_{\mathrm{GL}_n} = 1$ and $N_{\mathrm{GL}_n} = 1$. The Lie-good condition is consequently automatic for every prime in the general linear case.

For $r \geq 1$ define

$$U_r(\mathbf{G}) = \ker(\mathbf{G}(\mathbb{Z}_p) \to \mathbf{G}(\mathbb{Z}/p^r\mathbb{Z}))$$

and similarly $U_r(\mathbf{G}^{\mathrm{der}})$. Smoothness of the split reductive group scheme gives a canonical identification of additive groups

$$U_r(\mathbf{G})/U_{r+1}(\mathbf{G}) \simeq \mathfrak{g}_{\mathbb{F}_p}. \tag{2.6}$$

We recall this identification in a form which will also be used for square-zero coefficient rings. Let $I = p^r\mathbb{Z}/p^{r+1}\mathbb{Z}$. Since $I^2 = 0$, the kernel of $\mathbf{G}(\mathbb{Z}/p^{r+1}\mathbb{Z}) \to \mathbf{G}(\mathbb{Z}/p^r\mathbb{Z})$ is naturally the tangent space $\mathfrak{g}_{\mathbb{F}_p} \otimes I$. Thus an element $x \in U_r(\mathbf{G})$ has a well-defined leading term

$$\ell_r(x) \in \mathfrak{g}_{\mathbb{F}_p}.$$

Multiplication in the congruence quotient corresponds to addition of leading terms.

Two elementary calculations will be used repeatedly. If $x \in U_r(\mathbf{G})$ and $y \in U_s(\mathbf{G})$ have leading terms X and Y, then

$$[x, y] \in U_{r+s}(\mathbf{G}), \quad \text{and} \quad \ell_{r+s}([x, y]) = [X, Y]. \tag{2.7}$$

This may be checked after fixing any faithful integral representation $\mathbf{G} \hookrightarrow \mathrm{GL}_N$; the matrix computation is independent of the chosen embedding because it is the intrinsic Lie bracket on the tangent space. Likewise, for odd p,

$$x^p \in U_{r+1}(\mathbf{G}), \quad \text{and} \quad \ell_{r+1}(x^p) = X. \tag{2.8}$$

Indeed, after embedding in GL_N, write $x = 1 + p^r\widetilde{X} + O(p^{r+1})$. The binomial expansion gives

$$x^p = 1 + p^{r+1}\widetilde{X} + O(p^{r+2}).$$

For $2 \leq k \leq p-1$, the term $\binom{p}{k}(p^r\widetilde{X})^k$ has valuation at least $1 + kr \geq r + 2$, and the $k = p$ term has valuation $pr \geq r + 2$ because p is odd. The calculation also shows why $p = 2$ is excluded from the uniform statement.

If $D \subset \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$ is a closed subgroup, define

$$\Phi_r(D) = \{X \in \mathfrak{g}^{\mathrm{der}}_{\mathbb{F}_p} : \text{there exists } d \in D \cap U_r(\mathbf{G}^{\mathrm{der}}) \text{ with } \ell_r(d) = X\}. \tag{2.9}$$

Multiplication and taking integral powers show that $\Phi_r(D)$ is an $\mathbb{F}_p$-subspace. Equations (2.7) and (2.8) imply

$$[\Phi_r(D), \Phi_s(D)] \subset \Phi_{r+s}(D), \quad \text{and} \quad \Phi_r(D) \subset \Phi_{r+1}(D). \tag{2.10}$$

3. Finite-level deformation theory and relative obstruction classes

The relative lifting argument is the main point at which the present paper differs from the residual constructions in [10, 11, 12]. Throughout this section $\mathcal{G}$ denotes an arbitrary profinite group and $\mathbf{G}/\mathbb{Z}_p$ is a smooth affine group scheme; eventually $\mathcal{G} = G_{\{p\}}$ and $\mathbf{G}$ is our split reductive group.

Let $B' \twoheadrightarrow B$ be a surjection of finite local $\mathbb{Z}_p$-algebras with kernel I, and assume $I^2 = 0$. Smoothness gives surjectivity $\mathbf{G}(B') \twoheadrightarrow \mathbf{G}(B)$. The kernel is canonically the additive group $\mathfrak{g} \otimes I$. We denote by

$$\kappa_I : \mathfrak{g} \otimes I \longrightarrow \ker\big(\mathbf{G}(B') \to \mathbf{G}(B)\big)$$

the inverse of this canonical identification. Thus

$$\kappa_I(X)\kappa_I(Y) = \kappa_I(X + Y)$$

for $X, Y \in \mathfrak{g} \otimes I$. If $g \in \mathbf{G}(B)$ and $\widetilde{g} \in \mathbf{G}(B')$ is any lift, then conjugation by $\widetilde{g}$ on the kernel agrees with the adjoint action of g on $\mathfrak{g} \otimes I$. The action is independent of the choice of lift.

Let $\tau : \mathcal{G} \longrightarrow \mathbf{G}(B)$ be a continuous representation. We find that $\mathfrak{g} \otimes I$ is a continuous $\mathcal{G}$-module, which we denote $\operatorname{Ad}\tau(\mathfrak{g} \otimes I)$. Since the image of τ is finite, we may choose a set-theoretic section of $\mathbf{G}(B') \to \mathbf{G}(B)$ on that image and obtain a continuous map

$$\widetilde{\tau} : \mathcal{G} \longrightarrow \mathbf{G}(B')$$

lifting τ and satisfying $\widetilde{\tau}(1) = 1$. It need not be a homomorphism.

Lemma 3.1. *There is a canonically defined class*

$$o(\tau, B'/B) \in H^2(\mathcal{G}, \operatorname{Ad}\tau(\mathfrak{g} \otimes I))$$

which vanishes if and only if τ lifts to a continuous representation $\mathcal{G} \to \mathbf{G}(B')$.

Proof. For $g, h \in \mathcal{G}$, both $\widetilde{\tau}(g)\widetilde{\tau}(h)$ and $\widetilde{\tau}(gh)$ reduce to $\tau(gh)$ in $\mathbf{G}(B)$. Hence there is a unique $C(g, h) \in \mathfrak{g} \otimes I$ such that

$$\widetilde{\tau}(g)\widetilde{\tau}(h) = \kappa_I(C(g, h))\widetilde{\tau}(gh). \tag{3.1}$$

Associativity of multiplication in $\mathbf{G}(B')$, together with the additivity of the square-zero kernel, gives

$$C(g, h) + C(gh, k) = g \cdot C(h, k) + C(g, hk).$$

Thus C is a continuous 2-cocycle.

If $\widetilde{\tau}'$ is another lift, there is a unique continuous 1-cochain $b : \mathcal{G} \to \mathfrak{g} \otimes I$ with

$$\widetilde{\tau}'(g) = \kappa_I(b(g))\widetilde{\tau}(g).$$

Substitution into (3.1) shows that the new obstruction cocycle is $C + db$. Therefore the class $[C]$ is independent of all choices; this is $o(\tau, B'/B)$.

If the class vanishes, write $C = db$. Replacing $\widetilde{\tau}(g)$ by $\kappa_I(-b(g))\widetilde{\tau}(g)$ kills the obstruction and produces a genuine homomorphism. Conversely a genuine lift has zero obstruction. This proves the lemma. □

Remark 3.2. The calculation does not require $B' \to B$ to be a small extension in the Schlessinger sense. The only property used is $I^2 = 0$.

Lemma 3.3. *Suppose*

$$B' \twoheadrightarrow B'' \twoheadrightarrow B$$

are surjections of finite local rings. Put $I = \ker(B' \to B)$ and $I'' = \ker(B'' \to B)$ and assume $I^2 = 0$. Then the natural map $I \to I''$ carries $o(\tau, B'/B)$ to $o(\tau, B''/B)$.

Proof. Choose one set-theoretic lift of τ to $\mathbf{G}(B')$ and reduce it to $\mathbf{G}(B'')$. The multiplicative obstruction of the reduced lift is exactly the image of the original obstruction under $\mathfrak{g} \otimes I \to \mathfrak{g} \otimes I''$. Passing to cohomology gives the assertion. □

Fix now an integral representation

$$\rho_\infty : \mathcal{G} \longrightarrow \mathbf{G}(\mathbb{Z}_p)$$

and set

$$T = \operatorname{Ad}\rho_\infty(\mathfrak{g}_{\mathbb{Z}_p}), \qquad T_m = T/p^m T, \quad \text{and} \quad \rho_m = \rho_\infty \pmod{p^m}.$$

The following elementary calculation is the coefficient bookkeeping behind the relative method.

Lemma 3.4. *Let $R \geq 2$ and $j \geq R$. Set*

$$B' = \mathbb{Z}/p^{j+R+1}\mathbb{Z}, \qquad B = \mathbb{Z}/p^{j+1}\mathbb{Z}, \quad \textit{and} \quad I = p^{j+1}\mathbb{Z}/p^{j+R+1}\mathbb{Z}.$$

Then $I^2 = 0$, division by p^{j+1} identifies I additively with $\mathbb{Z}/p^R\mathbb{Z}$, and if $\tau : \mathcal{G} \to \mathbf{G}(B)$ satisfies $\tau \bmod p^R = \rho_R$, then

$$\operatorname{Ad}\tau(\mathfrak{g} \otimes I) \simeq T_R$$

as $\mathcal{G}$-modules.

Proof. Because $j \geq R$,

$$2(j+1) \geq j + R + 1,$$

so the product of two elements of I is divisible by p^{j+R+1} and hence $I^2 = 0$. Every element of the tangent kernel may be written uniquely as $p^{j+1}X$ with X defined modulo p^R. Conjugating by a lift of $\tau(g)$ gives

$$p^{j+1}X \longmapsto p^{j+1}\operatorname{Ad}(\widetilde{\tau(g)})(X).$$

After division by p^{j+1} and reduction modulo p^R, only $\tau(g) \bmod p^R = \rho_R(g)$ remains. Thus the action is the adjoint action through ρ_R, which is exactly T_R. □

We can now state the fixed-width lifting proposition. Its point is that the obstruction group at width R is allowed to be nonzero. Injectivity of one transition map is the required condition.

Proposition 3.5 (Relative lifting)**.** *Let $R \geq 2$ and suppose that*

$$H^2(\mathcal{G}, T_R) \longrightarrow H^2(\mathcal{G}, T_{R-1})$$

is injective. Let $j \geq R$, and suppose that a representation

$$\widetilde{\rho}_{j+R} : \mathcal{G} \longrightarrow \mathbf{G}(\mathbb{Z}/p^{j+R}\mathbb{Z})$$

has been constructed whose reduction modulo p^R is ρ_R. Let τ_{j+1} be its reduction modulo p^{j+1}. Then τ_{j+1} admits a lift

$$\widetilde{\rho}_{j+R+1} : \mathcal{G} \longrightarrow \mathbf{G}(\mathbb{Z}/p^{j+R+1}\mathbb{Z}).$$

Proof. Consider the square-zero extension

$$\mathbb{Z}/p^{j+R+1}\mathbb{Z} \longrightarrow \mathbb{Z}/p^{j+1}\mathbb{Z}.$$

By Lemma 3.4, its coefficient module for the obstruction to lifting τ_{j+1} is T_R. Denote the resulting class by

$$o_R(\tau_{j+1}) \in H^2(\mathcal{G}, T_R).$$

Now replace the upper ring by the intermediate quotient $\mathbb{Z}/p^{j+R}\mathbb{Z}$. The corresponding kernel has width $R-1$, so its coefficient module is T_{R-1}. Naturality of obstruction classes shows that the image of $o_R(\tau_{j+1})$ under

$$H^2(\mathcal{G}, T_R) \longrightarrow H^2(\mathcal{G}, T_{R-1})$$

is precisely the obstruction to lifting τ_{j+1} to level p^{j+R}. But such a lift is already given by $\widetilde{\rho}_{j+R}$. Hence the image of $o_R(\tau_{j+1})$ is zero. Injectivity forces $o_R(\tau_{j+1}) = 0$, and Lemma 3.1 gives the required lift. □

Remark 3.6. The new representation modulo p^{j+R+1} is not required to reduce to $\widetilde{\rho}_{j+R}$ modulo p^{j+R}. Only the mod p^{j+1} representation is fixed. This is the relative feature which replaces ordinary unobstructedness.

4. Cyclotomic descent and finite rational obstruction

We now specialize to $G = G_{\{p\}}$. The purpose of this section is only to prove that the rational obstruction $H^2(G_{\{p\}}, \mathbb{Q}_p(m))$ can be nonzero for at most finitely many integers m.

We let $Y := H^1(H, \mathbb{Q}_p/\mathbb{Z}_p)$ and $X := Y^\vee$. Since H acts trivially on $\mathbb{Q}_p/\mathbb{Z}_p$, one has

$$H^1(H, \mathbb{Q}_p/\mathbb{Z}_p) = \mathrm{Hom}_{\mathrm{cts}}(H, \mathbb{Q}_p/\mathbb{Z}_p).$$

Every homomorphism in this group factors through the maximal abelian pro-p quotient of H. Consequently, by Pontryagin duality, X is naturally identified with this maximal abelian pro-p quotient.

Equivalently, let M_∞ denote the maximal abelian pro-p extension of F_∞ contained in $\mathbb{Q}_{\{p\}}$. Then M_∞/F_∞ is unramified at every finite prime not lying above p, and

$$X \simeq \mathrm{Gal}(M_\infty/F_\infty).$$

Thus X is the standard p-ramified Iwasawa module attached to the cyclotomic extension $F_\infty/\mathbb{Q}$. The action of $\Gamma = \mathrm{Gal}(F_\infty/\mathbb{Q})$ on H by conjugation induces an action on X, making X a compact module over the Iwasawa algebra $\Lambda = \mathbb{Z}_p[[\Gamma_1]]$, and more precisely over $\Lambda[\Delta_p]$. It is finitely generated over Λ; see [8, Proposition 11.3.1]. We have that

$$H^2(H, \mathbb{Q}_p/\mathbb{Z}_p) = 0, \tag{4.1}$$

cf. [8, Theorem 10.3.25]. Since $\chi|_H = 1$, the same vanishing holds after every Tate twist.

For $m \in \mathbb{Z}$, recall that

$$e_m = \frac{1}{p-1} \sum_{\delta \in \Delta_p} \omega(\delta)^{-m} \delta \in \mathbb{Z}_p[\Delta_p]$$

and set

$$f_m = \gamma^{-1} - u^{-m} \in \Lambda.$$

We note that $f_m = T - (u^{-m} - 1)$. In particular (f_m) is a height-one prime and $\Lambda/(f_m) \simeq \mathbb{Z}_p$.

Proposition 4.1. *For every $m \in \mathbb{Z}$ there is a natural isomorphism*

$$H^2\big(G_{\{p\}}, (\mathbb{Q}_p/\mathbb{Z}_p)(m)\big)^\vee \simeq e_m X[f_m]. \tag{4.2}$$

Proof. Put $A := \mathbb{Q}_p/\mathbb{Z}_p$ and $A_m := A(m)$. Recall that $H = \mathrm{Gal}(\mathbb{Q}_{\{p\}}/F_\infty)$ and $\Gamma = \mathrm{Gal}(F_\infty/\mathbb{Q})$. Since F_∞ contains all p-power roots of unity, the cyclotomic character is trivial on H. Thus H acts trivially on the Tate-twist factor of A_m. Consequently, for every $q \geq 0$, there is an identification $H^q(H, A_m) = H^q(H, A)(m)$ as Γ-modules.

We first apply the Hochschild–Serre spectral sequence to the exact sequence

$$1 \longrightarrow H \longrightarrow G_{\{p\}} \longrightarrow \Gamma \longrightarrow 1.$$

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$$E_2^{a,b} = H^a\big(\Gamma, H^b(H, A_m)\big) \Longrightarrow H^{a+b}(G_{\{p\}}, A_m).$$

We are interested in total degree 2. There are therefore three potential terms, namely $E_2^{0,2}$, $E_2^{1,1}$ and $E_2^{2,0}$. The first of these vanishes. Indeed, weak Leopoldt (4.1) for the cyclotomic extension gives $H^2(H, A) = 0$. Since the Tate twist does not change the action of H, it follows that $H^2(H, A_m) = 0$, and hence $E_2^{0,2} = 0$.

The third term also vanishes. Recall that $\Gamma = \Delta_p \times \Gamma_1$, where $\Gamma_1 \simeq \mathbb{Z}_p$ and $|\Delta_p| = p-1$. The group Γ_1 has p-cohomological dimension one, while a finite group of order prime to p has p-cohomological dimension zero. Hence $\mathrm{cd}_p(\Gamma) = 1$. Since A_m is a discrete p-primary module, this gives $E_2^{2,0} = H^2(\Gamma, A_m) = 0$.

Thus the only possibly nonzero term in total degree 2 is

$$E_2^{1,1} = H^1\big(\Gamma, H^1(H, A_m)\big).$$

Since $H^1(H, A_m) = Y(m)$, this term is $H^1(\Gamma, Y(m))$. Moreover, no differential can enter $E_2^{1,1}$, since its possible source would have negative first index, and no differential can leave it: the first possible target is $E_2^{3,0}$, which vanishes because $\mathrm{cd}_p(\Gamma) = 1$. Therefore $E_\infty^{1,1} = E_2^{1,1}$. As all the other graded pieces of the Hochschild–Serre filtration on $H^2(G_{\{p\}}, A_m)$ vanish, we obtain a canonical isomorphism

$$H^2(G_{\{p\}}, A_m) \simeq H^1(\Gamma, Y(m)). \tag{4.3}$$

We next compute the group on the right. Apply Hochschild–Serre to the exact sequence $1 \to \Gamma_1 \to \Gamma \to \Delta_p \to 1$. Since $|\Delta_p| = p-1$ is prime to p, one has $H^a(\Delta_p, D) = 0$ for every $a > 0$ and every discrete p-primary Δ_p-module D. Indeed, multiplication by $|\Delta_p|$ is an automorphism of D, so the usual averaging operator makes the invariants functor exact. It follows that

$$H^1(\Gamma, Y(m)) \simeq H^1(\Gamma_1, Y(m))^{\Delta_p}.$$

Let γ be the fixed topological generator of Γ_1 and put $u = \chi(\gamma)$. For any discrete p-primary Γ_1-module D, continuous cohomology of $\Gamma_1 \simeq \mathbb{Z}_p$ gives $H^1(\Gamma_1, D) = D/(\gamma - 1)D$. On the twisted module $Y(m)$, the element γ acts as $u^m\gamma$, where the second γ denotes its original action on Y. Consequently,

$$H^1(\Gamma_1, Y(m)) \simeq Y/(u^m\gamma - 1)Y.$$

For brevity, denote this quotient by Q_m. Thus $H^1(\Gamma, Y(m)) = Q_m^{\Delta_p}$.

We now dualize. Since $X = Y^\vee$, the Pontryagin dual of Q_m is the annihilator in X of $(u^m\gamma - 1)Y$. Under the usual contragredient action on the Pontryagin dual, the adjoint of γ acting on Y is γ^{-1} acting on X. Therefore

$$Q_m^\vee = \ker\big(u^m\gamma^{-1} - 1 : X \longrightarrow X\big).$$

Multiplying the operator by the unit u^{-m} does not change its kernel. Hence

$$Q_m^\vee = \ker(\gamma^{-1} - u^{-m} : X \longrightarrow X) = X[f_m],$$

where, by definition, $f_m = \gamma^{-1} - u^{-m}$.

It remains to keep track of the action of the finite group Δ_p. This is the point at which the idempotent e_m enters. For $\delta \in \Delta_p$, the action on the twisted module $Y(m)$ is the original action on Y multiplied by $\chi(\delta)^m = \omega(\delta)^m$. Therefore an element of Q_m is fixed by Δ_p after twisting precisely when, with respect to the original action on Y, it belongs to the ω^{-m}-eigenspace. In terms of the standard idempotents, this says

$$Q_m^{\Delta_p} = e_{-m}Q_m.$$

Pontryagin duality reverses the finite character. More explicitly, if y transforms under Δ_p by ω^{-m}, then the contragredient action on a functional pairing nontrivially with y is by ω^m. Equivalently, under the involution $\delta \mapsto \delta^{-1}$ on $\mathbb{Z}_p[\Delta_p]$, the idempotent e_{-m} is carried to e_m. It follows that

$$(Q_m^{\Delta_p})^\vee = (e_{-m}Q_m)^\vee \simeq e_m Q_m^\vee.$$

Using $Q_m^\vee = X[f_m]$ and the fact that f_m commutes with the Δ_p-action, we finally obtain

$$H^1(\Gamma, Y(m))^\vee \simeq e_m X[f_m].$$

Combining this with (4.3) proves (4.2). □

Lemma 4.2. *Let p be odd and let T be a finite free $\mathbb{Z}_p$-module equipped with a continuous action of $G_{\{p\}}$. Then*

$$H^3(G_{\{p\}}, T) = 0.$$

Proof. Write $T = \varprojlim_r T/p^rT$. By [8, Theorem 2.7.5], continuous cohomology with compact coefficients gives a natural exact sequence

$$0 \longrightarrow \varprojlim_r{}^1 H^2(G_{\{p\}}, T/p^rT) \longrightarrow H^3(G_{\{p\}}, T) \longrightarrow \varprojlim_r H^3(G_{\{p\}}, T/p^rT) \longrightarrow 0.$$

Here $\varprojlim^1$ denotes the first right-derived functor of the inverse limit. For every $r \geq 1$, the module T/p^rT is a finite discrete p-primary $G_{\{p\}}$-module, and $H^2(G_{\{p\}}, T/p^rT)$ is finite; see [8, Theorem 8.3.19]. For each fixed r, the images of the groups at higher levels in $H^2(G_{\{p\}}, T/p^rT)$ form a descending chain of subgroups of a finite group and hence eventually stabilize. Thus the inverse system satisfies the Mittag–Leffler condition, and [8, Proposition 2.7.4] gives

$$\varprojlim_r{}^1 H^2(G_{\{p\}}, T/p^rT) = 0.$$

Since p is odd, $\mathrm{cd}_p(G_{\{p\}}) = 2$. Therefore

$$H^3(G_{\{p\}}, T/p^rT) = 0$$

for every $r \geq 1$. Both outer terms in the preceding exact sequence vanish, and hence

$$H^3(G_{\{p\}}, T) = 0.$$

□

Lemma 4.3. *For every $m \in \mathbb{Z}$, the groups $H^i(G_{\{p\}}, \mathbb{Z}_p(m))$ are finitely generated $\mathbb{Z}_p$-modules for $i = 1, 2$, and the canonical maps*

$$H^i(G_{\{p\}}, \mathbb{Z}_p(m)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \longrightarrow H^i(G_{\{p\}}, \mathbb{Q}_p(m)) \qquad \text{for} \quad i = 1, 2$$

are isomorphisms. In particular, the degree-two map

$$H^2(G_{\{p\}}, \mathbb{Z}_p(m)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \longrightarrow H^2(G_{\{p\}}, \mathbb{Q}_p(m))$$

is an isomorphism. Moreover

$$\dim_{\mathbb{Q}_p} H^2(G_{\{p\}}, \mathbb{Q}_p(m)) = \mathrm{corank}_{\mathbb{Z}_p} H^2\big(G_{\{p\}}, (\mathbb{Q}_p/\mathbb{Z}_p)(m)\big).$$

Proof. By [8, Theorem 8.3.19], $H^2(G_{\{p\}}, \mathbb{F}_p(m))$ is finite. By Lemma 4.2, we have

$$H^3(G_{\{p\}}, \mathbb{Z}_p(m)) = 0.$$

Hence the coefficient sequence

$$0 \longrightarrow \mathbb{Z}_p(m) \xrightarrow{p} \mathbb{Z}_p(m) \longrightarrow \mathbb{F}_p(m) \longrightarrow 0$$

gives

$$H^2(G_{\{p\}}, \mathbb{Z}_p(m))/pH^2(G_{\{p\}}, \mathbb{Z}_p(m)) \simeq H^2(G_{\{p\}}, \mathbb{F}_p(m)).$$

Thus the reduction modulo p of the compact $\mathbb{Z}_p$-module $H^2(G_{\{p\}}, \mathbb{Z}_p(m))$ is finite. By the topological Nakayama lemma, it follows that $H^2(G_{\{p\}}, \mathbb{Z}_p(m))$ is finitely generated over $\mathbb{Z}_p$. The same coefficient sequence gives an injection

$$H^1(G_{\{p\}}, \mathbb{Z}_p(m))/pH^1(G_{\{p\}}, \mathbb{Z}_p(m)) \hookrightarrow H^1(G_{\{p\}}, \mathbb{F}_p(m)).$$

The group on the right is finite by [8, Theorem 8.3.19], so another application of the topological Nakayama lemma gives finite generation of $H^1(G_{\{p\}}, \mathbb{Z}_p(m))$.

Continuous cohomology in this finite cohomological-dimension setting commutes with passage from a $\mathbb{Z}_p$-lattice to its associated $\mathbb{Q}_p$-vector space. Equivalently, passing to the direct limit through the lattices

$$\mathbb{Z}_p \cdot p^{-k}(m) \hookrightarrow \mathbb{Z}_p \cdot p^{-k-1}(m)$$

gives

$$H^2(G_{\{p\}}, \mathbb{Z}_p(m)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \simeq H^2(G_{\{p\}}, \mathbb{Q}_p(m)).$$

The identical argument in degree one gives

$$H^1(G_{\{p\}}, \mathbb{Z}_p(m)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \simeq H^1(G_{\{p\}}, \mathbb{Q}_p(m)).$$

For this compatibility, see also [8, Chapter 2, §7].

For the corank assertion, use

$$0 \longrightarrow \mathbb{Z}_p(m) \longrightarrow \mathbb{Q}_p(m) \longrightarrow (\mathbb{Q}_p/\mathbb{Z}_p)(m) \longrightarrow 0.$$

By Lemma 4.2, the resulting long exact sequence ends with

$$H^2(G_{\{p\}}, \mathbb{Z}_p(m)) \longrightarrow H^2(G_{\{p\}}, \mathbb{Q}_p(m)) \longrightarrow H^2(G_{\{p\}}, (\mathbb{Q}_p/\mathbb{Z}_p)(m)) \longrightarrow 0.$$

The image of the first term is a full $\mathbb{Z}_p$-lattice in the finite-dimensional $\mathbb{Q}_p$-space in the middle. The quotient therefore has $\mathbb{Z}_p$-corank equal to the dimension of that $\mathbb{Q}_p$-space. □

The next commutative-algebra observation is what makes the exceptional set finite.

Lemma 4.4. *Let M be a finitely generated $\Lambda = \mathbb{Z}_p[[T]]$-module, and let $f = T - a$ with $a \in p\mathbb{Z}_p$. If $M[f]$ has positive $\mathbb{Z}_p$-rank, then (f) is an associated prime of M.*

Proof. Since $M[f]$ has positive $\mathbb{Z}_p$-rank, there exists $x \in M[f]$ which is not $\mathbb{Z}_p$-torsion. In particular, x is not annihilated by any power of p.

Since $fx = 0$, the action of Λ on the cyclic submodule Λx factors through $\Lambda/(f)$. Because $a \in p\mathbb{Z}_p$, evaluation at $T = a$ gives an isomorphism $\Lambda/(f) \simeq \mathbb{Z}_p$. Thus Λx may be regarded as a $\mathbb{Z}_p$-module generated by x.

We claim that $\mathrm{Ann}_\Lambda(x) = (f)$. The inclusion $(f) \subseteq \mathrm{Ann}_\Lambda(x)$ is immediate from $fx = 0$. Conversely, suppose that $g \in \Lambda$ satisfies $gx = 0$, and let $\bar{g}$ denote its image in $\Lambda/(f) \simeq \mathbb{Z}_p$. Then $\bar{g}x = 0$. If $\bar{g} \neq 0$, write $\bar{g} = p^r u$ with $r \geq 0$ and $u \in \mathbb{Z}_p^\times$. Since multiplication by u is an automorphism of the $\mathbb{Z}_p$-module Λx, the equality $\bar{g}x = 0$ implies $p^r x = 0$, contradicting the choice of x. Hence $\bar{g} = 0$, so $g \in (f)$.

Therefore $\mathrm{Ann}_\Lambda(x) = (f)$. Since $\Lambda/(f) \simeq \mathbb{Z}_p$ is an integral domain, (f) is prime, and hence, by definition, $(f) \in \mathrm{Ass}_\Lambda(M)$. □

Proposition 4.5. *There is a finite set $\mathcal{E}_p \subset \mathbb{Z}$ such that*

$$H^2(G_{\{p\}}, \mathbb{Q}_p(m)) = 0 \tag{4.4}$$

for every $m \notin \mathcal{E}_p$. Moreover $H^2(G_{\{p\}}, \mathbb{Q}_p) = 0$.

Proof. By Proposition 4.1 and Lemma 4.3, a nonzero rational group in (4.4) implies that $e_m X[f_m]$ has positive $\mathbb{Z}_p$-rank. Lemma 4.4 therefore shows that (f_m) is an associated prime of the finitely generated module $e_m X$. Note that e_m depends only on m modulo $p-1$. Thus there are $p-1$ modules $e_m X$ to consider and for each module, the set $\mathrm{Ass}_\Lambda(e_m X)$ is finite; see [7, Theorem 6.5(i)]. Finally the ideals (f_m) are pairwise distinct because u has infinite order in $1+p\mathbb{Z}_p$. Thus only finitely many integers m can occur.

The trivial twist requires a separate argument. Since $\mathbb{Q}_p$ has trivial $G_{\{p\}}$-action, one has $H^0(G_{\{p\}}, \mathbb{Q}_p) = \mathbb{Q}_p$. Moreover,

$$H^1(G_{\{p\}}, \mathbb{Q}_p) = \mathrm{Hom}_{\mathrm{cts}}(G_{\{p\}}, \mathbb{Q}_p).$$

By global class field theory, the maximal abelian pro-p quotient of $G_{\{p\}}$ has $\mathbb{Z}_p$-rank one. Indeed, its unique infinite $\mathbb{Z}_p$-extension is the cyclotomic $\mathbb{Z}_p$-extension of $\mathbb{Q}$. Hence $\dim_{\mathbb{Q}_p} H^1(G_{\{p\}}, \mathbb{Q}_p) = 1$.

We now apply the global Euler–Poincaré characteristic formula for p-adic representations; see [4, Lemma 9.7]. For a finite-dimensional $\mathbb{Q}_p$-representation V of $G_{\{p\}}$, one has

$$\sum_{i=0}^{2} (-1)^i \dim_{\mathbb{Q}_p} H^i(G_{\{p\}}, V) = \dim_{\mathbb{Q}_p} V^{c=1} - \dim_{\mathbb{Q}_p} V,$$

where c denotes complex conjugation.

For the trivial representation $V = \mathbb{Q}_p$, complex conjugation acts trivially, so the right-hand side is zero. Since $H^0(G_{\{p\}}, \mathbb{Q}_p)$ and $H^1(G_{\{p\}}, \mathbb{Q}_p)$ both have dimension one, it follows that

$$H^2(G_{\{p\}}, \mathbb{Q}_p) = 0.$$

□

5. The integral toral representation

Let $L = X_*(\mathbf{T})$. By hypothesis there is $\lambda_0 \in L$ such that $\langle \alpha, \lambda_0 \rangle$ is odd for every $\alpha \in \Delta$. Put $\mathcal{B} = \Delta \cup (-\Delta)$.

Lemma 5.1 (Choice of cocharacter)**.** *Let $\mathcal{E} \subset \mathbb{Z}$ be finite. There exists a cocharacter $\lambda \in X_*(\mathbf{T})$ such that:*

1. *$\langle \alpha, \lambda \rangle$ is odd for every $\alpha \in \Delta$;*
2. *$\langle \beta, \lambda \rangle \notin \mathcal{E}$ for every $\beta \in \Phi$;*
3. *the integers $\langle \alpha, \lambda \rangle$, for $\alpha \in \mathcal{B}$, are pairwise distinct;*
4. *if $\dim Z(\mathbf{G}) = 1$, the image $\lambda^{\mathrm{ab}} \in X_*(\mathbf{G}/\mathbf{G}^{\mathrm{der}})$ is nonzero.*

Proof. Put $L = X_*(\mathbf{T})$ and begin with the coset $\lambda_0 + 2L$, where λ_0 is the odd cocharacter fixed above. Every element of this coset has the same simple-root pairings modulo 2 as λ_0. Hence condition (1) holds automatically.

We next impose conditions (2) and (3). Fix a root $\beta \in \Phi$ and an integer $e \in \mathcal{E}$. The condition $\langle \beta, \lambda \rangle = e$ defines a proper affine hyperplane in $L \otimes_{\mathbb{Z}} \mathbb{Q}$. Indeed, the root β is a nonzero character of $\mathbf{T}$, and hence the linear functional $\lambda \mapsto \langle \beta, \lambda \rangle$ is nonzero on $L \otimes_{\mathbb{Z}} \mathbb{Q}$.

Likewise, if $\alpha, \beta \in \mathcal{B}$ are distinct, then the condition $\langle \alpha, \lambda \rangle = \langle \beta, \lambda \rangle$ is equivalent to $\langle \alpha - \beta, \lambda \rangle = 0$. Since $\alpha \neq \beta$ as characters of $\mathbf{T}$, the character $\alpha - \beta$ is nonzero, and this again defines a proper hyperplane in $L \otimes_{\mathbb{Z}} \mathbb{Q}$.

There are only finitely many roots, finitely many elements of $\mathcal{E}$, and finitely many pairs of elements of $\mathcal{B}$. Consequently the cocharacters which fail either condition (2) or condition (3)

lie in a finite union of proper affine hyperplanes. A finite union of proper affine hyperplanes cannot contain the full-rank lattice coset $\lambda_0 + 2L$. We may therefore choose λ in this coset which satisfies conditions (1)–(3). In fact, there are infinitely many such choices.

Suppose now that $\dim Z(\mathbf{G}) = 1$. Since $\mathbf{G}$ is split, the connected center $Z(\mathbf{G})^0$ is a one-dimensional split torus. Choose a nonzero cocharacter $\zeta \in X_*(Z(\mathbf{G})^0)$. The natural map $Z(\mathbf{G})^0 \to \mathbf{G}/\mathbf{G}^{\mathrm{der}}$ has finite kernel, namely $Z(\mathbf{G})^0 \cap \mathbf{G}^{\mathrm{der}}$. It follows that the image $\zeta^{\mathrm{ab}} \in X_*(\mathbf{G}/\mathbf{G}^{\mathrm{der}})$ is nonzero. Since every root of $\mathbf{G}$ is trivial on the center, replacing λ by $\lambda + 2N\zeta$ does not change any of the root pairings $\langle \beta, \lambda \rangle$. Thus conditions (1)–(3) remain unchanged.

The lattice $X_*(\mathbf{G}/\mathbf{G}^{\mathrm{der}})$ has rank one. Hence $\lambda^{\mathrm{ab}} + 2N\zeta^{\mathrm{ab}}$ can vanish for at most one integer N. Choosing any other N gives condition (4), while preserving conditions (1)–(3). □

Apply Lemma 5.1 with $\mathcal{E} = \mathcal{E}_p \cup \{0\}$. Fix the resulting cocharacter λ and define

$$\rho_\infty = \lambda \circ \chi : G_{\{p\}} \longrightarrow \mathbf{T}(\mathbb{Z}_p) \subset \mathbf{G}(\mathbb{Z}_p). \tag{5.1}$$

For every root β put

$$d_\beta = \langle \beta, \lambda \rangle.$$

Then $d_{-\beta} = -d_\beta$, every simple-root weight is odd, and every root weight lies outside $\mathcal{E}_p$. Put $T = \mathrm{Ad}\, \rho_\infty(\mathfrak{g}_{\mathbb{Z}_p})$ and $T_r = T/p^r T$. By (2.5), we have that

$$T = \mathfrak{t}_{\mathbb{Z}_p} \oplus \bigoplus_{\beta \in \Phi} \mathbb{Z}_p(d_\beta) X_\beta. \tag{5.2}$$

Proposition 5.2. *The group $H^2(G_{\{p\}}, T)$ is finite.*

Proof. By (5.2), we have a decomposition of $G_{\{p\}}$-modules

$$T = \mathfrak{t}_{\mathbb{Z}_p} \oplus \bigoplus_{\beta \in \Phi} \mathbb{Z}_p(d_\beta) X_\beta,$$

where the action on $\mathfrak{t}_{\mathbb{Z}_p}$ is trivial. Thus $\mathfrak{t}_{\mathbb{Z}_p}$ is a finite direct sum of copies of $\mathbb{Z}_p$.

Applying Lemma 4.3 to the trivial twist and to each of the finitely many twists d_β, we find that $H^2(G_{\{p\}}, T)$ is a finitely generated $\mathbb{Z}_p$-module and that

$$H^2(G_{\{p\}}, T) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \simeq H^2\big(G_{\{p\}}, T \otimes_{\mathbb{Z}_p} \mathbb{Q}_p\big).$$

After tensoring (5.2) with $\mathbb{Q}_p$, the toral summand is a finite direct sum of copies of the trivial representation $\mathbb{Q}_p$, while the root summand indexed by β is $\mathbb{Q}_p(d_\beta)X_\beta$. Proposition 4.5 gives

$$H^2(G_{\{p\}}, \mathbb{Q}_p) = 0$$

and, since $d_\beta \notin \mathcal{E}_p$ for every $\beta \in \Phi$,

$$H^2(G_{\{p\}}, \mathbb{Q}_p(d_\beta)) = 0.$$

Hence

$$H^2\big(G_{\{p\}}, T \otimes_{\mathbb{Z}_p} \mathbb{Q}_p\big) = 0.$$

It follows that

$$H^2(G_{\{p\}}, T) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p = 0.$$

Therefore $H^2(G_{\{p\}}, T)$ is a finitely generated torsion $\mathbb{Z}_p$-module, and hence is finite. □

Choose $e \geq 0$ such that

$$p^e H^2(G_{\{p\}}, T) = 0. \tag{5.3}$$

Proposition 5.3. *For every $r \geq e+1$, the natural reduction map*

$$H^2(G_{\{p\}}, T_r) \longrightarrow H^2(G_{\{p\}}, T_{r-1})$$

is an isomorphism.

Proof. It follows from Lemma 4.2 that $H^3(G_{\{p\}}, T) = 0$. Consider the short exact sequence

$$0 \longrightarrow T \xrightarrow{p^r} T \longrightarrow T_r \longrightarrow 0.$$

Since $H^3(G_{\{p\}}, T) = 0$, the corresponding long exact sequence contains

$$H^2(G_{\{p\}}, T) \xrightarrow{p^r} H^2(G_{\{p\}}, T) \longrightarrow H^2(G_{\{p\}}, T_r) \longrightarrow 0.$$

It follows that

$$H^2(G_{\{p\}}, T_r) \simeq H^2(G_{\{p\}}, T)/p^r H^2(G_{\{p\}}, T). \tag{5.4}$$

The reduction map $T_r \to T_{r-1}$ is compatible with the two short exact sequences defining T_r and T_{r-1}. Under the identifications (5.4), the induced map on cohomology is therefore the natural quotient map

$$H^2(G_{\{p\}}, T)/p^r H^2(G_{\{p\}}, T) \longrightarrow H^2(G_{\{p\}}, T)/p^{r-1} H^2(G_{\{p\}}, T).$$

By (5.3), multiplication by p^e annihilates $H^2(G_{\{p\}}, T)$. If $r \geq e+1$, then both p^r and p^{r-1} annihilate this group. Hence both the source and target of the preceding quotient map identify canonically with $H^2(G_{\{p\}}, T)$, and under these identifications the transition map is the identity. It is therefore an isomorphism. □

Fix from now on an integer

$$R \geq \max\{2, e+1\}. \tag{5.5}$$

Thus

$$H^2(G_{\{p\}}, T_R) \longrightarrow H^2(G_{\{p\}}, T_{R-1})$$

is injective, which is the only obstruction-theoretic property needed in the relative tower.

We next construct the root-valued cohomology classes used at the initial finite level.

Lemma 5.4. *Let d be odd and suppose $H^2(G_{\{p\}}, \mathbb{Q}_p(d)) = 0$. Then*

$$\dim_{\mathbb{Q}_p} H^1(G_{\{p\}}, \mathbb{Q}_p(d)) = 1.$$

Moreover $H^1(G_{\{p\}}, \mathbb{Z}_p(d))$ has $\mathbb{Z}_p$-rank one and contains an integral class mapping to a nonzero rational class.

Proof. Since $d \neq 0$, $H^0(G_{\{p\}}, \mathbb{Q}_p(d)) = 0$. Complex conjugation acts on $\mathbb{Q}_p(d)$ by $(-1)^d = -1$. The global Euler–Poincare characteristic formula gives

$$\dim H^0 - \dim H^1 + \dim H^2 = -1.$$

The first and third terms vanish, and therefore $\dim H^1 = 1$. The degree-one assertion of Lemma 4.3 identifies $H^1(G_{\{p\}}, \mathbb{Z}_p(d)) \otimes \mathbb{Q}_p$ with this one-dimensional rational space. The integral group therefore has rank one, and any element mapping to a nonzero rational vector gives the required class. □

Lemma 5.5. *If $d \neq 0$, the restriction map*

$$H^1(G_{\{p\}}, \mathbb{Q}_p(d)) \longrightarrow H^1(H, \mathbb{Q}_p(d))$$

is injective.

Proof. Inflation–restriction for

$$1 \longrightarrow H \longrightarrow G_{\{p\}} \xrightarrow{\chi} \mathbb{Z}_p^\times \longrightarrow 1$$

identifies the kernel with $H^1(\mathbb{Z}_p^\times, \mathbb{Q}_p(d))$. Write $\mathbb{Z}_p^\times = \Delta_p \times (1+p\mathbb{Z}_p)$. Since $|\Delta_p|$ is invertible in $\mathbb{Q}_p$, its positive-degree cohomology vanishes. A topological generator of $1+p\mathbb{Z}_p$ acts on $\mathbb{Q}_p(d)$ by $u^d \neq 1$, so

$$H^1(1+p\mathbb{Z}_p, \mathbb{Q}_p(d)) = \mathbb{Q}_p/(u^d-1)\mathbb{Q}_p = 0.$$

Hence the kernel of restriction is zero. □

For each $\alpha \in \mathcal{B}$, the integer d_α is odd and lies outside $\mathcal{E}_p$. Choose an integral cocycle

$$z_\alpha \in Z^1(G_{\{p\}}, \mathbb{Z}_p(d_\alpha)) \tag{5.6}$$

representing a nonzero rational class. By Lemma 5.5, its restriction to H is nonzero. Because H acts trivially on the coefficient module, the restriction is an ordinary continuous homomorphism $H \longrightarrow \mathbb{Z}_p$. Its image is a nonzero closed subgroup of $\mathbb{Z}_p$, hence equals $p^{s_\alpha}\mathbb{Z}_p$ for a unique $s_\alpha \geq 0$. Set

$$s = \max_{\alpha \in \mathcal{B}} s_\alpha. \tag{5.7}$$

6. The initial finite-level lift

We now construct a representation modulo a sufficiently large power of p whose image already contains the positive and negative simple-root directions. Define

$$F(g) = \sum_{\alpha \in \mathcal{B}} z_\alpha(g) X_\alpha. \tag{6.1}$$

Each summand lies in the root line $\mathbb{Z}_p(d_\alpha)X_\alpha$, so F is a 1-cocycle with values in $T = \mathrm{Ad}\,\rho_\infty(\mathfrak{g}_{\mathbb{Z}_p})$.

Choose $v = 1+p$ and choose $\sigma \in G_{\{p\}}$ with $\chi(\sigma) = v$. For $\alpha \in \mathcal{B}$ put

$$\lambda_\alpha = v^{d_\alpha} \in \mathbb{Z}_p^\times.$$

By the choice of λ, the integers d_α for $\alpha \in \mathcal{B}$ are pairwise distinct; because v has infinite order, the p-adic units λ_α are pairwise distinct. Conjugation by $\rho_\infty(\sigma)$ acts on X_α as multiplication by λ_α.

There are finitely many pairs $\alpha \neq \beta$, so define

$$m = 1 + \max_{\substack{\alpha,\beta \in \mathcal{B} \\ \alpha \neq \beta}} v_p(\lambda_\alpha - \lambda_\beta). \tag{6.2}$$

Then

$$v_p(\lambda_\alpha - \lambda_\beta) < m \qquad (\alpha \neq \beta). \tag{6.3}$$

If $A \geq m$, the eigenvalues remain pairwise distinct modulo p^A. Set

$$C = m(|\mathcal{B}| - 1). \tag{6.4}$$

Lemma 6.1. *Let $A \geq m$ and let $M \subset T_A$ be a $G_{\{p\}}$-stable $\mathbb{Z}/p^A\mathbb{Z}$-submodule. Suppose*

$$x = \sum_{\beta \in \mathcal{B}} c_\beta X_\beta \in M$$

is supported on the simple-root lines. Fix $\alpha \in \mathcal{B}$ and suppose $c_\alpha \neq 0$ with $v_p(c_\alpha) \leq s_0$. If $s_0 + C < A$, then

$$p^{s_0+C} X_\alpha \in M.$$

Proof. Let

$$D = \operatorname{Ad}\rho_A(\sigma)$$

and define

$$P_\alpha(X) = \prod_{\substack{\beta\in\mathcal{B}\\ \beta\neq\alpha}} (X - \lambda_\beta).$$

Because M is stable under the Galois action, it is stable under D and hence under every polynomial in D with coefficients in $\mathbb{Z}/p^A\mathbb{Z}$. On the root vectors, $D(X_\beta) = \lambda_\beta X_\beta$. Therefore $P_\alpha(D)X_\beta = 0$ for $\beta \neq \alpha$, while

$$P_\alpha(D)X_\alpha = \prod_{\beta\neq\alpha} (\lambda_\alpha - \lambda_\beta)X_\alpha.$$

It follows that

$$P_\alpha(D)x = c_\alpha \prod_{\beta\neq\alpha} (\lambda_\alpha - \lambda_\beta)X_\alpha.$$

Write $t = v_p(c_\alpha)$ and

$$r_\alpha = v_p\left(\prod_{\beta\neq\alpha} (\lambda_\alpha - \lambda_\beta)\right).$$

There are $|\mathcal{B}| - 1$ factors, each of valuation strictly less than m, hence $r_\alpha < C$. Consequently $t + r_\alpha < s_0 + C < A$, so the displayed coefficient is nonzero modulo p^A and may be written $p^{t+r_\alpha}u_\alpha$ with u_α a unit. Multiplying by $u_\alpha^{-1}p^{s_0+C-t-r_\alpha}$ gives the desired element. □

Choose A so large that

$$A \geq m, \qquad A > s + C + R + 2. \tag{6.5}$$

If $\dim Z(\mathbf{G}) = 1$, we also enlarge A so that the reduction of the nonzero central character $\lambda^{\mathrm{ab}}\circ\chi$ modulo p^A already has nontrivial pro-p image. The ideal $p^A\mathbb{Z}/p^{2A}\mathbb{Z}$ is square-zero. Therefore smoothness gives a canonical additive isomorphism

$$\kappa_A : T_A \xrightarrow{\sim} \ker\big(\mathbf{G}(\mathbb{Z}/p^{2A}\mathbb{Z}) \to \mathbf{G}(\mathbb{Z}/p^A\mathbb{Z})\big). \tag{6.6}$$

We record explicitly how this square-zero identification interacts with congruence depth.

Lemma 6.2. *Let $0 \leq t < A$ and let $X \in \mathfrak{g}_{\mathbb{Z}_p}$ have nonzero reduction $\bar{X} \in \mathfrak{g}_{\mathbb{F}_p}$. Then $\kappa_A(p^tX \bmod p^A)$ lies in the image of the congruence subgroup $U_{A+t}(\mathbf{G})$ modulo p^{2A}, and its leading term at depth $A+t$ is $\bar{X}$.*

Proof. This is the functorial description of the tangent kernel for the square-zero ideal $p^A\mathbb{Z}/p^{2A}\mathbb{Z}$. After choosing a faithful integral representation $\mathbf{G} \hookrightarrow \mathrm{GL}_N$, the element $\kappa_A(p^tX)$ has the form

$$1 + p^{A+t}\widetilde{X} \pmod{p^{2A}},$$

where $\widetilde{X} \bmod p = \bar{X}$. Hence it is trivial modulo p^{A+t}, nontrivial in general modulo p^{A+t+1}, and its image in $U_{A+t}(\mathbf{G})/U_{A+t+1}(\mathbf{G})$ is exactly $\bar{X}$. The conclusion is independent of the chosen faithful representation because the leading term is the intrinsic tangent vector. □

We define

$$\rho_{2A}^*(g) = \kappa_A(F(g) \bmod p^A)\,\rho_{2A}(g). \tag{6.7}$$

This formula is the intrinsic group-scheme replacement for $(1 + p^AF(g))\rho_\infty(g)$ in the matrix case.

Lemma 6.3. *The map*

$$\rho_{2A}^* : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}/p^{2A}\mathbb{Z})$$

is a continuous representation and its reduction modulo p^A is ρ_A.

Proof. Continuity is immediate. The cocycle relation for F is

$$F(gh) = F(g) + \mathrm{Ad}(\rho_\infty(g))F(h).$$

Modulo p^A the same identity holds in T_A. The multiplication law in the square-zero kernel and the conjugation formula give

$$\begin{aligned}\rho_{2A}^*(g)\rho_{2A}^*(h) &= \kappa_A(F(g))\rho_{2A}(g)\kappa_A(F(h))\rho_{2A}(h)\\ &= \kappa_A(F(g) + \mathrm{Ad}(\rho_A(g))F(h))\,\rho_{2A}(gh)\\ &= \kappa_A(F(gh))\rho_{2A}(gh)\\ &= \rho_{2A}^*(gh).\end{aligned}$$

No quadratic error remains because the kernel is square-zero. Finally $\kappa_A(F(g))$ lies in the kernel of reduction to level A, so the reduction of ρ_{2A}^* is ρ_A. □

Put $K_A = \ker \rho_A$. Since ρ_{2A}^* reduces to ρ_A, its image on K_A lies in the square-zero congruence kernel. Hence there is a unique additive subgroup $M_A \subset T_A$ such that

$$\rho_{2A}^*(K_A) = \kappa_A(M_A). \tag{6.8}$$

An additive subgroup of the finite $\mathbb{Z}/p^A\mathbb{Z}$-module T_A is stable under integer multiplication and is therefore a $\mathbb{Z}/p^A\mathbb{Z}$-submodule.

Lemma 6.4. *The submodule $M_A \subset T_A$ is stable under the adjoint action of $G_{\{p\}}$ through ρ_A.*

Proof. Take $X \in M_A$ and choose $h \in K_A$ with $\rho_{2A}^*(h) = \kappa_A(X)$. For $g \in G_{\{p\}}$, normality of K_A gives $ghg^{-1} \in K_A$. Thus $\rho_{2A}^*(ghg^{-1}) = \kappa_A(Y)$ for some $Y \in M_A$. On the other hand,

$$\rho_{2A}^*(ghg^{-1}) = \rho_{2A}^*(g)\kappa_A(X)\rho_{2A}^*(g)^{-1}.$$

Conjugation on the square-zero kernel depends only on the reduction of $\rho_{2A}^*(g)$ modulo p^A, which is $\rho_A(g)$. Hence

$$Y = \mathrm{Ad}(\rho_A(g))X.$$

Therefore M_A is Galois-stable. □

We can now isolate every positive and negative simple root.

Proposition 6.5. *Put*

$$q = A + s + C. \tag{6.9}$$

For every $\alpha \in \mathcal{B}$, the image of ρ_{2A}^* contains an element lying in the q-th congruence subgroup whose leading term at level q is X_α. Moreover

$$q + 1 \leq 2A - R. \tag{6.10}$$

Proof. Fix $\alpha \in \mathcal{B}$. By definition of s_α choose $h_\alpha \in H$ with

$$z_\alpha(h_\alpha) = p^{s_\alpha}u_\alpha, \qquad u_\alpha \in \mathbb{Z}_p^\times.$$

Since $\chi(h_\alpha) = 1$, one has $\rho_\infty(h_\alpha) = 1$, so $H \subset K_A$. Formula (6.7) then becomes

$$\rho_{2A}^*(h_\alpha) = \kappa_A(F(h_\alpha) \bmod p^A).$$

Thus $F(h_\alpha) \bmod p^A \in M_A$. The vector $F(h_\alpha)$ is supported on the simple-root lines and its X_α-coefficient has valuation exactly s_α. Since M_A is Galois-stable, Lemma 6.1 applies and gives

$$p^{s_\alpha + C} X_\alpha \in M_A.$$

Multiplying by p^{s-s_α} gives $p^{s+C}X_\alpha \in M_A$. Under the identification of the square-zero kernel, Lemma 6.2 says that the finite-level image contains an element whose first nonzero congruence term occurs at depth $A + s + C = q$ and equals X_α.

Finally, $A > s + C + R + 2$ implies

$$q + 1 = A + s + C + 1 \leq 2A - R.$$

The root data therefore lie strictly below the lower truncation at which the relative lifting tower will begin. □

7. Proof of the main theorem

We now lift the finite-level representation of the preceding section to characteristic zero.

Recall that R was chosen so that

$$H^2(G_{\{p\}}, T_R) \longrightarrow H^2(G_{\{p\}}, T_{R-1})$$

is injective. Set

$$j_0 = 2A - R. \tag{7.1}$$

The choice of A gives $j_0 > A > R$. Moreover ρ^*_{2A} reduces modulo p^A to ρ_A, and hence modulo p^R to ρ_R.

Proposition 7.1. *For every $j \geq j_0$ there exist continuous representations*

$$\tau_j : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}/p^j\mathbb{Z})$$

and

$$\varrho_{j+R} : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}/p^{j+R}\mathbb{Z})$$

such that

(1) $\varrho_{j+R} \pmod{p^j} = \tau_j$;
(2) $\tau_{j+1} \pmod{p^j} = \tau_j$;
(3) $\varrho_{j+R} \pmod{p^R} = \rho_R$;
(4) $\tau_{j_0} = \rho^*_{2A} \pmod{p^{j_0}}$.

Proof. At the initial stage put $\varrho_{j_0+R} = \rho^*_{2A}$ and define τ_{j_0} by reduction. Since $j_0 + R = 2A$, properties (1), (3), and (4) hold.

Suppose now that τ_j and ϱ_{j+R} have been constructed. Define τ_{j+1} to be the reduction of ϱ_{j+R} modulo p^{j+1}. Then $\tau_{j+1} \pmod{p^j} = \tau_j$ and $\tau_{j+1} \pmod{p^R} = \rho_R$. Proposition 3.5 applies with the fixed width R and gives a representation

$$\varrho_{j+R+1} : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}/p^{j+R+1}\mathbb{Z})$$

whose reduction modulo p^{j+1} is τ_{j+1}. Reducing further modulo p^R gives ρ_R. This completes the induction. □

The high-level representations ϱ_{j+R} are auxiliary and need not be compatible. The compatibility is only in the lower truncations τ_j.

Corollary 7.2. *There exists a continuous representation*

$$\rho : G_{\{p\}} \longrightarrow \mathbf{G}(\mathbb{Z}_p)$$

whose reduction modulo p^j is τ_j for every $j \geq j_0$.

Proof. The compatible system in Proposition 7.1(2) defines a homomorphism to $\varprojlim_j \mathbf{G}(\mathbb{Z}/p^j\mathbb{Z}) = \mathbf{G}(\mathbb{Z}_p)$. Continuity follows from the continuity of the finite quotients. □

By Proposition 6.5, for every $\alpha \in \mathcal{B}$ there is $g_\alpha \in G_{\{p\}}$ such that the finite-level element $\rho_{2A}^*(g_\alpha)$ has leading term X_α at depth q, and $q+1 \leq j_0$.

Corollary 7.3. *For every $\alpha \in \mathcal{B}$ there exists $g_\alpha \in G_{\{p\}}$ such that*

$$\rho(g_\alpha) \in U_q(\mathbf{G}) \quad \textit{and} \quad \ell_q(\rho(g_\alpha)) = X_\alpha.$$

Proof. The final representation ρ and ρ_{2A}^* have the same reduction modulo p^{j_0}. Since $q+1 \leq j_0$, they have the same reduction through the level at which the q-th leading term is measured. Hence each simple-root leading term constructed in Proposition 6.5 survives unchanged in ρ. □

Put $\mathcal{H} = \rho(G_{\{p\}})$ and $\mathcal{D} = \overline{[\mathcal{H}, \mathcal{H}]}$. Every commutator in $\mathbf{G}(\mathbb{Z}_p)$ lies in $\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$, so $\mathcal{D} \subset \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$.

Let $\bar{g}$ generate $\mathbb{F}_p^\times$. Since the mod-p cyclotomic character is surjective, choose $\sigma_0 \in G_{\{p\}}$ with $\overline{\chi}(\sigma_0) = \bar{g}$ and put $t = \rho(\sigma_0)$. The reduction of ρ modulo p agrees with that of the toral representation ρ_∞, because $\rho \pmod{p^R} = \rho_R$. Thus conjugation by t on the α-root line modulo p is multiplication by $\bar{g}^{d_\alpha}$.

Lemma 7.4. *For every $\alpha \in \mathcal{B}$, $X_\alpha \in \Phi_q(\mathcal{D})$.*

Proof. Choose $x_\alpha = \rho(g_\alpha)$ as in Corollary 7.3. The commutator $[t, x_\alpha]$ belongs to $\mathcal{D}$ and lies in $U_q(\mathbf{G}^{\mathrm{der}})$. Its leading term is

$$(\bar{g}^{d_\alpha} - 1)X_\alpha.$$

For $\alpha \in \mathcal{B}$, the integer d_α is odd. Since the order $p-1$ of $\bar{g}$ is even, $p-1$ cannot divide d_α. Hence $\bar{g}^{d_\alpha} \neq 1$, so the displayed scalar is nonzero in $\mathbb{F}_p$. Because $\Phi_q(\mathcal{D})$ is an $\mathbb{F}_p$-subspace, division by this nonzero scalar yields $X_\alpha \in \Phi_q(\mathcal{D})$. □

This argument is worth noting: it uses only the parity of the simple-root weights and therefore imposes no lower bound on p beyond oddness.

Let $h_{\max}$ be the maximum height of a positive root. Starting from $X_{\pm\alpha}$ for the simple roots, the Chevalley relations generate every root vector. We include the depth bookkeeping because it is needed to place all directions in one common congruence layer.

Lemma 7.5. *Let $\beta \in \Phi$. Then*

$$X_\beta \in \Phi_{|\,\mathrm{ht}(\beta)|q}(\mathcal{D}).$$

Proof. We argue first for positive roots by induction on $h = \mathrm{ht}(\beta)$. If $h = 1$, the root is simple and the assertion is Lemma 7.4. Suppose $h > 1$. There is a simple root $\alpha \in \Delta$ such that $\gamma = \beta - \alpha$ is a positive root of height $h-1$. By induction,

$$X_\alpha \in \Phi_q(\mathcal{D}), \quad \text{and} \quad X_\gamma \in \Phi_{(h-1)q}(\mathcal{D}).$$

The commutator property gives

$$[X_\alpha, X_\gamma] \in \Phi_{hq}(\mathcal{D}).$$

By the Chevalley relation,

$$[X_\alpha, X_\gamma] = N_{\alpha,\gamma} X_\beta.$$

The Lie-good hypothesis says that $N_{\alpha,\gamma}$ is nonzero modulo p. Since $\Phi_{hq}(\mathcal{D})$ is an $\mathbb{F}_p$-space, division by this scalar gives X_β at the required level. The negative roots are handled in exactly the same way using the negative simple roots. □

For every simple root α we also have

$$[X_\alpha, X_{-\alpha}] = H_\alpha.$$

Hence

$$H_\alpha \in \Phi_{2q}(\mathcal{D}) \quad \text{for} \quad \alpha \in \Delta. \tag{7.2}$$

By the Lie-good hypothesis, the reductions of all root vectors and these simple coroot directions span $\mathfrak{g}^{\mathrm{der}}_{\mathbb{F}_p}$.

Put

$$r_0 = \max\{2, h_{\max}\}q. \tag{7.3}$$

Repeated use of the p-power inclusion $\Phi_r(\mathcal{D}) \subset \Phi_{r+1}(\mathcal{D})$ moves every root vector and every H_α from the level at which it was produced to the common level r_0. We have proved:

Proposition 7.6. *One has*

$$\Phi_{r_0}(\mathcal{D}) = \mathfrak{g}^{\mathrm{der}}_{\mathbb{F}_p}.$$

Lemma 7.7. *Let $D \subset \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$ be a closed subgroup. Suppose that for some $r \geq 1$ the natural map*

$$D \cap U_r(\mathbf{G}^{\mathrm{der}}) \longrightarrow U_r(\mathbf{G}^{\mathrm{der}})/U_{r+1}(\mathbf{G}^{\mathrm{der}}) \simeq \mathfrak{g}^{\mathrm{der}}_{\mathbb{F}_p}$$

is surjective. Then

$$U_r(\mathbf{G}^{\mathrm{der}}) \subset D.$$

Proof. First, surjectivity persists at every deeper level. Suppose it holds at level $s \geq r$ and take a prescribed class $X \in \mathfrak{g}^{\mathrm{der}}_{\mathbb{F}_p}$ at level $s+1$. Choose $x \in D \cap U_s(\mathbf{G}^{\mathrm{der}})$ with leading term X. By (2.8), x^p belongs to $D \cap U_{s+1}(\mathbf{G}^{\mathrm{der}})$ and has the same leading term X at level $s+1$. Induction gives surjectivity at every level.

Now fix $u \in U_r(\mathbf{G}^{\mathrm{der}})$. Construct elements $P_s \in D$ such that

$$uP_s^{-1} \in U_s(\mathbf{G}^{\mathrm{der}}).$$

Start with $P_r = 1$. If P_s has been constructed, put $E_s = uP_s^{-1}$. Choose $d_s \in D \cap U_s(\mathbf{G}^{\mathrm{der}})$ with the same class as E_s in U_s/U_{s+1}. Then $E_s d_s^{-1} \in U_{s+1}$. Set

$$P_{s+1} = d_s P_s.$$

Since $P_{s+1}^{-1} = P_s^{-1} d_s^{-1}$,

$$uP_{s+1}^{-1} = E_s d_s^{-1} \in U_{s+1}.$$

Thus $P_s \to u$. Closedness of D gives $u \in D$. Since u was arbitrary, $U_r(\mathbf{G}^{\mathrm{der}}) \subset D$. □

Proposition 7.8. *The representation ρ constructed above satisfies*

$$U_{r_0}(\mathbf{G}^{\mathrm{der}}) \subset \mathcal{D} \subset \rho(G_{\{p\}}).$$

Proof. Proposition 7.6 says exactly that $\mathcal{D} \cap U_{r_0}(\mathbf{G}^{\mathrm{der}})$ surjects onto the first congruence quotient at depth r_0. Lemma 7.7 therefore gives $U_{r_0}(\mathbf{G}^{\mathrm{der}}) \subset \mathcal{D}$. The second inclusion is immediate from the definition of the closed commutator subgroup. □

We first record two elementary topological facts used in passing from the derived group to the full reductive group.

Lemma 7.9. *Let p be odd and let C be a closed subgroup of the one-dimensional split torus $\mathbf{G}_m(\mathbb{Z}_p) = \mathbb{Z}_p^\times$. If the projection of C to the pro-p factor $1 + p\mathbb{Z}_p$ is nontrivial, then C is open in $\mathbb{Z}_p^\times$.*

Proof. For odd p one has

$$\mathbb{Z}_p^\times = \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

The p-adic logarithm identifies $1 + p\mathbb{Z}_p$ topologically with the additive group $p\mathbb{Z}_p$, and every nonzero closed subgroup of $p\mathbb{Z}_p$ is of the form $p^a\mathbb{Z}_p$ for some $a \geq 1$. Thus the pro-p projection of C is open in $1 + p\mathbb{Z}_p$. Since the complementary factor μ_{p-1} is finite, C has finite index in $\mathbb{Z}_p^\times$ and is therefore open. □

Lemma 7.10. *Let $H \subset \mathbf{G}(\mathbb{Z}_p)$ be a closed subgroup. Suppose that $H \cap \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$ is open in $\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$ and that the image of H in $(\mathbf{G}/\mathbf{G}^{\mathrm{der}})(\mathbb{Z}_p)$ is open. Then H is open in $\mathbf{G}(\mathbb{Z}_p)$.*

Proof. Let $q : \mathbf{G}(\mathbb{Z}_p) \to (\mathbf{G}/\mathbf{G}^{\mathrm{der}})(\mathbb{Z}_p)$ be the quotient map and put $V = q(H)$, which is open by hypothesis. Then $q^{-1}(V)$ is open in $\mathbf{G}(\mathbb{Z}_p)$. If $g \in q^{-1}(V)$, choose $h \in H$ with $q(h) = q(g)$. Then $h^{-1}g \in \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$, so

$$q^{-1}(V) = H\,\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p).$$

Consequently

$$[q^{-1}(V) : H] \leq [\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p) : H \cap \mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)] < \infty.$$

Thus H has finite index in the open subgroup $q^{-1}(V)$, and hence H is open in $\mathbf{G}(\mathbb{Z}_p)$. □

We now prove the main theorem.

Proof of Theorem 1.1. If $\mathbf{G}$ is semisimple, Proposition 7.8 already shows that $\rho(G_{\{p\}})$ is open in $\mathbf{G}(\mathbb{Z}_p)$. Suppose now that $\dim Z(\mathbf{G}) = 1$ and put $\mathbf{S} = \mathbf{G}/\mathbf{G}^{\mathrm{der}}$. The final representation ρ agrees with the initial finite-level representation modulo p^{j_0}, and the latter agrees with ρ_∞ modulo p^A. Hence the image of ρ in $\mathbf{S}(\mathbb{Z}_p)$ has the same reduction modulo p^A as the nonzero cyclotomic character $\lambda^{\mathrm{ab}} \circ \chi$. By the additional choice of A in Section 6, this reduction already has nontrivial pro-p image. A closed subgroup of the one-dimensional p-adic torus with nontrivial pro-p image is open by Lemma 7.9. Thus the image of ρ in the central quotient is open. Together with the principal congruence subgroup in $\mathbf{G}^{\mathrm{der}}(\mathbb{Z}_p)$, Lemma 7.10 proves that $\rho(G_{\{p\}})$ is open in $\mathbf{G}(\mathbb{Z}_p)$.

It remains to obtain infinitely many representations. The construction works for every sufficiently large choice of the width parameter R: if R is enlarged, the cocharacter λ, the finite group $H^2(G_{\{p\}}, T)$, the cocycles z_α, and the constants s and C remain fixed; one simply chooses A still larger so that (6.5) holds. The resulting characteristic-zero representation satisfies

$$\rho^{(R)} \equiv \rho_\infty \pmod{p^R}.$$

Every open-image representation $\rho^{(R)}$ differs from the toral representation ρ_∞, because the latter has image contained in $\mathbf{T}(\mathbb{Z}_p)$. Suppose finitely many distinct representations $\rho^{(R_1)}, \ldots, \rho^{(R_m)}$ have already been chosen. For each i, choose a level b_i at which $\rho^{(R_i)} \not\equiv \rho_\infty \pmod{p^{b_i}}$. Choose a new width $R > \max_i b_i$ and carry out the construction. Then the new representation is congruent to ρ_∞ modulo every p^{b_i}, so it cannot equal any of the previous representations. Induction produces infinitely many pairwise distinct representations.

In fact, the same construction gives infinitely many $\mathbf{G}(\mathbb{Z}_p)$-conjugacy classes, which is the stronger and more natural form of the infinitude statement. Suppose that $\rho_1, \ldots, \rho_m$ have already been chosen in pairwise distinct $\mathbf{G}(\mathbb{Z}_p)$-conjugacy classes. For each i consider, for $b \geq 1$,

$$\mathcal{C}_{i,b} = \left\{ g \in \mathbf{G}(\mathbb{Z}_p) : g\rho_i g^{-1} \equiv \rho_\infty \pmod{p^b} \right\}.$$

Each $\mathcal{C}_{i,b}$ is closed in the compact group $\mathbf{G}(\mathbb{Z}_p)$, and the sets are nested as b increases. If $\mathcal{C}_{i,b}$ were nonempty for every b, compactness would give an element in their intersection. Such an element would conjugate ρ_i to ρ_∞ exactly. This is impossible, because ρ_i has open derived image whereas ρ_∞ has image in the torus $\mathbf{T}(\mathbb{Z}_p)$. Hence for each i there is an integer b_i such that $\mathcal{C}_{i,b_i} = \varnothing$.

Choose the next width R larger than every b_i. The newly constructed representation $\rho^{(R)}$ satisfies $\rho^{(R)} \equiv \rho_\infty \pmod{p^R}$, and therefore cannot be conjugate to any ρ_i: otherwise a conjugating element would belong to $\mathcal{C}_{i,b_i}$. Induction produces infinitely many pairwise nonconjugate representations. □

We finish by checking every hypothesis in the general theorem for $\mathbf{G} = \mathrm{GL}_n$.

Proof of Corollary1.2. Take the diagonal torus. Its cocharacter lattice is $\mathbb{Z}^n$, and for

$$\lambda(t) = \mathrm{diag}(t^{a_1}, \dots, t^{a_n})$$

the root $e_i - e_j$ has weight $a_i - a_j$. Let $\mathcal{E}_p$ be the finite exceptional set from Proposition 4.5. We choose $a_1, \dots, a_n$ recursively so that

(1) $a_i \equiv i \pmod 2$;
(2) $a_i - a_j \notin \mathcal{E}_p \cup \{0\}$ for $i \neq j$;
(3) all ordered differences $a_i - a_j$, $i \neq j$, are pairwise distinct;
(4) $a_1 + \cdots + a_n \neq 0$.

At the $(r+1)$-st stage the prescribed parity leaves an infinite arithmetic progression of possible values for a_{r+1}. Avoiding a root weight in the finite exceptional set excludes finitely many values. Equality of a new difference with an old difference gives a linear equation in a_{r+1} and therefore excludes one value. Equality between two new ordered differences gives either an equality already forced by their indices or an equation of the form $2a_{r+1} = a_i + a_j$, again excluding at most one value. Thus only finitely many integers of the prescribed parity are forbidden. At the last stage we also avoid the single value which would make the total sum zero. This proves the required choice for every n and every odd p.

The adjacent differences

$$d_i = a_i - a_{i+1}$$

are odd. Hence the positive and negative simple roots satisfy the odd cocharacter condition. All root weights avoid $\mathcal{E}_p$, so the rational obstruction vanishing used in the general proof holds. The nonzero sum $a_1 + \cdots + a_n$ says that the determinant of the toral representation is the nonzero cyclotomic power

$$\det \rho_\infty = \chi^{a_1 + \cdots + a_n},$$

which has open image in $\mathbb{Z}_p^\times$.

It remains only to check the Lie-good condition. In the standard matrix realization take

$$X_{e_i - e_j} = e_{ij}.$$

Whenever i, j, k are distinct,

$$[e_{ij}, e_{jk}] = e_{ik},$$

so every nonzero root-string structure constant needed to generate the higher roots is ± 1. These constants remain invertible modulo every prime. Opposite roots satisfy

$$[e_{ij}, e_{ji}] = e_{ii} - e_{jj}.$$

The off-diagonal matrix units together with

$$e_{11} - e_{22}, \ldots, e_{n-1,n-1} - e_{nn}$$

span $\mathfrak{sl}_n(\mathbb{F}_p)$ for every prime p. This remains true when $p \mid n$: the trace-zero diagonal space still has dimension $n-1$ and is spanned by the adjacent differences. Thus no prime is excluded by the Chevalley or coroot calculations.

Finally, the p-power congruence calculation used in (2.8) requires only that p be odd. No step of the proof compares p with n, and no class-group or irregularity condition remains. Theorem 1.1 therefore applies to GL_n for every odd prime p and every $n > 1$, proving Corollary 1.2. □

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